A CMS Open Data Search for $H \to \tau\tau$ in the $\mu\tau_h$ Final State

Analysis my_analysis[1]

[1] *CMS Open Data reduced-sample analysis*

(Dated: June 12, 2026)

A reduced CMS 2012 Open Data search for Higgs boson decays to tau pairs is performed in the $\mu\tau_h$ final state. The final result is a validated visible-mass template fit in VBF, boosted, and zero-jet categories. The fit finds $\mu = 0.438^{+4.944}_{-0.438}$ and sets an observed 95% CLs upper limit $\mu < 10.764$, with median expected limit $\mu < 10.738$. The result is a low-sensitivity Open Data limit, not evidence for a Standard Model-strength signal.

Data and method. The analysis uses the public Run2012B/C TauPlusX reduced samples and available ggH, VBF, DY, ttbar, and W+jets simulation at $L = 11.467\ \text{fb}^{-1}$. Events pass the $\mu + \tau_h$ trigger, tight muon and hadronic-tau selections, opposite-sign charge, and $m_T(\mu, E_T^{\text{miss}}) < 40$ GeV. The categories are VBF, boosted, and zero-jet. The likelihood is a binned HistFactory model with common signal-strength modifier μ and CLs limits computed with pyhf.

Background model. The W+jets scale is derived in a high-m_T control region: $S_W = 0.853 \pm 0.037$. VBF-category MC backgrounds receive a VBF-like control scale $S_{\text{VBF}} = 0.557 \pm 0.051$. Reducible QCD/fake contamination is estimated from same-sign low-m_T data after non-QCD subtraction. The systematic model includes luminosity, tau/open-data acceptance, DY/Z normalization, W and VBF control terms, same-sign QCD transfer, and MC statistical uncertainties.

Validation and result. The combined visible-mass validation gives data/background 0.981 and $\chi^2/\text{ndf} = 0.722$. The VBF category carries the largest local residual with max pull 2.66. The optimized classifier study has better expected sensitivity but fails its observed validation gate (fail) and is not used as a final result.

Interpretation. CMS Run 1 and later CMS publications observe or provide evidence for $H \to \tau\tau$ with many channels and a fuller calibration program. The present result is consistent only as a low-sensitivity public-data exercise: its median expected limit of $\mu < 10.738$ is an order of magnitude above the Standard Model rate.

TABLE I. Final visible-mass category yields. Background includes MC backgrounds and the same-sign QCD/fake estimate.

Category	Data	Bkg.	Signal	χ^2/ndf
VBF	78	96.3	1.59	1.45
Boosted	2183	2364.4	9.17	0.61
Zero-jet	8451	8456.8	14.34	0.11

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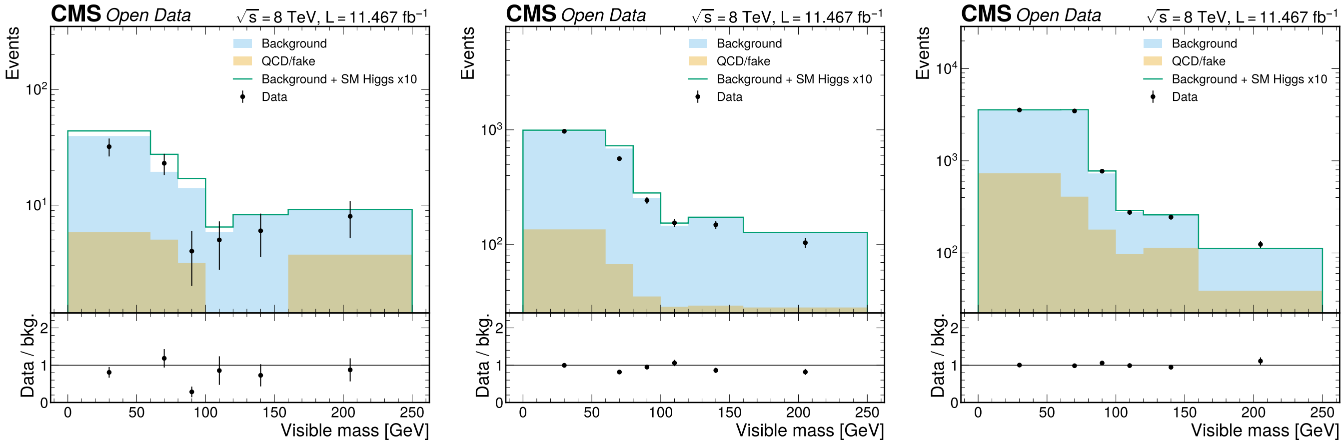

FIG. 1. Visible-mass validation in the final fit categories: VBF (left), boosted (center), and zero-jet (right). These plots define the validated model used for the final limit.

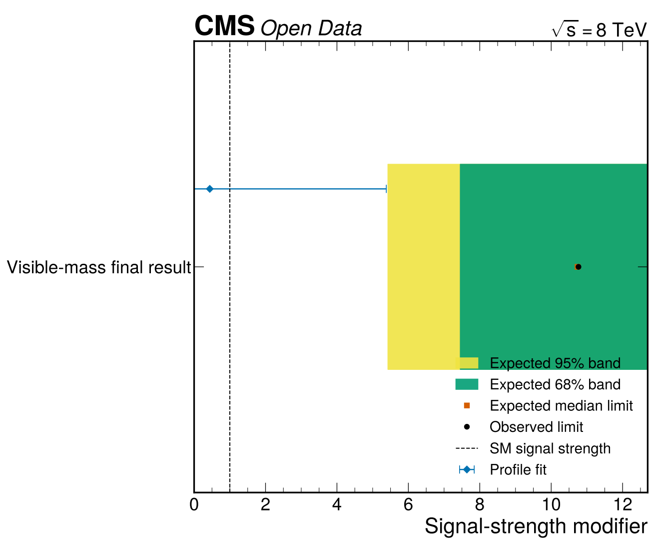

FIG. 2. Final visible-mass CLs limit summary. The expected band and observed marker show that the reduced Open Data analysis has no Standard Model resolving power.